REVERSE ENGINEERING · APPLE POWER ACCESSORIES

Apple USB-C and MagSafe 3 power path

Host controllers, cable firmware, and the UARP transport

Three firmware objects separated, three cable releases extracted, and the cross-version facts established without modifying any device.

REPORT	VERSION	DATE	CLASSIFICATION
RR-2026-1114	1.0	2026-11-11	Internal
TARGET	BASELINE	PREPARED BY	TEMPLATE
3.2.0	3.1.8	Sein	v1.2.0

DOCUMENT CONTROL

REPORT	Apple USB-C and MagSafe 3 power path
REPORT ID	RR-2026-1114
VERSION	1.0
DATE	2026-11-11
CLASSIFICATION	Internal
BASELINE	MagSafe 3 · 3.1.8
TARGET	MagSafe 3 · 3.2.0
PREPARED BY	Sein
SOURCE MATERIAL	Apple public UARP assets, hpmdiagnose register dumps, IORegistry captures
RENDERER	Typst · Research Reports template 1.2.0

CONTENTS

1 Executive summary 3
2 Method 3
3 Claim register 4
4 Corrections 5
5 Limitations 6

1 Executive summary

Read-only investigation of the power path on an Apple Silicon MacBook Pro: the USB-C PD controllers, the MagSafe 3 cable, and the host update stack. Three firmware objects are involved and are routinely conflated; they were separated, and the cable's three public releases were extracted and compared at byte and function level.

VERDICT

The host power path, the UARP container format and the cross-version byte facts are established. No host-observable behavioural difference between cable releases is demonstrated, and semantic equivalence remains `OPEN`.

KEY FINDINGS

01 **C01 — observed** — The charger negotiates 20 V at 3.390 A (67.80 W) on the MagSafe port.

02 **C04 — reproduced** — The UARP SuperBinary container is a 44-byte header with 40-byte payload records; dataOffset plus dataLength reconciles i

03 **C06 — reproduced** — The power-negotiation table content (5/9/15/20 V at 3.00 A) is identical in all builds while its address moves.

04 **C12 — observed** — A 14-case mov-pc dispatch table with 6 unique targets and 8 of 14 cases to one target is present in both 3.1.x builds an

23	9	0
CLAIMS RECONCILED	MANDATORY REGRESSIONS	UNRESOLVED CONTRADICTIONS

2 Method

Evidence came from IORegistry and IOAccessoryManager captures, a deterministic `hpmdiagnose` register dump, Apple's own public firmware manifests, and static

analysis of the extracted images. Labels are one scale: `OBSERVED`, `REPRODUCED`, `INFERRED`, `UNVERIFIED`, `REFUTED`, `OPEN`.

SCOPE

All hardware interaction was read-only. No firmware was flashed, no update was triggered, no accessory command was sent. Proprietary images are not redistributed with this report; hashes and acquisition notes are.

---

3 Claim register

Every claim below is checked across all eleven research documents by `scripts/verify_claims.py`, which fails on a missing statement, a conflicting label in the same paragraph, a withdrawn phrase in a conclusion, or a broken evidence reference.

- **C01** [OBSERVED] The charger negotiates 20 V at 3.390 A (67.80 W) on the MagSafe port.
- **C02** [OBSERVED] Four HPM controllers are present at RID 0, 1, 2 and 5; RID 5 carries the MagSafe port.
- **C03** [OBSERVED] HPM controller firmware FW003.115.00, the MagSafe cable at 3.2.0 and the adapter value 1030030 are three distinct objects, not revisions of one image.
- **C04** [REPRODUCED] The UARP SuperBinary container is a 44-byte header with 40-byte payload records; dataOffset plus dataLength reconciles in all four files.
- **C05** [OBSERVED] Three MagSafe 3 releases were acquired and each SHA-1 matches Apple’s manifest _Measurement.
- **C06** [REPRODUCED] The power-negotiation table content (5/9/15/20 V at 3.00 A) is identical in all builds while its address moves.
- **C07** [OBSERVED] Exactly one Apple SVID literal changes across the three releases: 0x05AC0002 disappears and 0x05AC0000 gains an occurrence.
- **C08** [REPRODUCED] Between 3.1.12 and 3.2.0 the compared bodies share 11 non-trivial literals, lose 7 and gain 1; 28 is present only in 3.2.0.
- **C09** [REFUTED] The vector word at 0x0001111d points to 0x1111c, which is constant data and not a reset handler.
- **C10** [OBSERVED] The 140 W control image’s pointers 0xec89 and 0xec71 exceed its 0xea00 payload, which rules out a base-zero mapping only; a non-zero base such as 0x400 would place them inside.
- **C11** [REPRODUCED] The frame header formula (sop<<4)|ceil(bytes/4) reproduces the observed 0x21 for SOP selector 2 and a 4-byte payload; the meaning of the sop field outside the observed selector set is open.
- **C12** [OBSERVED] A 14-case mov-pc dispatch table with 6 unique targets and 8 of 14 cases to one target is present in both 3.1.x builds and is not detected in 3.2.0.
- **C12b** [OPEN] The 15-case and 13-case tables in 3.2.0 are candidates only; no correspondence to the 3.1.x switch is shown.
- **C13** [REFUTED] Claims that 3.2.0 replaced or reorganised the dispatcher, or is a rebuild rather than a patch, are withdrawn.

- **C14** [UNVERIFIED] The Apple SVID literal sites are not established as a UVDM builder; that identification remains unverified.
- **C15** [REFUTED] The 5.9 % / 73.1 % aligned-window overlap figures measure layout; the inference from them to a rewrite is refuted.
- **C16** [REPRODUCED] Ghidra function coverage is 38.8 / 39.4 / 43.3 % of image bytes, and the switch and literal sites lie outside the analysed set.
- **C17** [REFUTED] Hard-reset handling change is refuted: Hard_Reset has no MessageType, so a 9-22 message-type switch cannot speak to it.
- **C18** [OPEN] No soft-to-hard reset escalation ladder was found; the question stays open across all three images.
- **C19** [OBSERVED] 70 W adapter firmware was not found by three enumeration methods (mesu path probes, Pallas lookup with a control type, and a broad type sweep); this is a scoped negative, not proof of nonexistence.
- **C20** [OPEN] Signature enforcement below the manifest hash, anti-rollback, and whether the Mode 7 read verb can touch flash are all open.
- **C21** [OBSERVED] hpmdiagnose reads the full HPM register file unprivileged; access is entitlement-gated, not root-gated.
- **C22** [OPEN] Semantic equivalence of 3.2.0 to 3.1.12 is undeterminable at the achieved coverage and is open.

4 Corrections

The independent audit found arithmetic and labelling errors; each is corrected at the data level, not re-worded.

TABLE 1

Corrected and withdrawn claims

CLAIM	STATUS	CORRECTED STATEMENT
C06	REPRODUCED	The power-negotiation table content (5/9/15/20 V at 3.00 A) is identical in all builds while its address moves.
C08	REPRODUCED	Between 3.1.12 and 3.2.0 the compared bodies share 11 non-trivial literals, lose 7 and gain 1; 28 is present only in 3.2.0.
C09	REFUTED	The vector word at 0x0001111d points to 0x1111c, which is constant data and not a reset handler.
C10	OBSERVED	The 140 W control image’s pointers 0xec89 and 0xec71 exceed its 0xea00 payload, which rules out a base-zero mapping only; a non-zero base such as 0x40
C13	REFUTED	Claims that 3.2.0 replaced or reorganised the dispatcher, or is a rebuild rather than a patch, are withdrawn.
C15	REFUTED	The 5.9 % / 73.1 % aligned-window overlap figures measure layout; the inference from them to a rewrite is refuted.

CLAIM	STATUS	CORRECTED STATEMENT
C16	REPRODUCED	Ghidra function coverage is 38.8 / 39.4 / 43.3 % of image bytes, and the switch and literal sites lie outside the analysed set.
C17	REFUTED	Hard-reset handling change is refuted: Hard_Reset has no MessageType, so a 9-22 message-type switch cannot speak to it.

Source: `DIFF.md` §1 and §4a, `DISPOSITION.md`; each row re-derived from the extracted images or the pseudocode.

5 Limitations

- Function coverage is 38.8 / 39.4 / 43.3 % of the three images, and the control-message switch and Apple SVID literal sites sit outside the analysed set.
- Per-function matching is not reported: both calibrated runs were void, and the causes are recorded rather than hidden.
- The 70 W adapter firmware is not publicly obtainable, so adapter behaviour is host-observed only.
- The load base for the accessory images, the C99 header fields, and any signature or anti-rollback enforcement below the manifest hash are `OPEN`.

The unproven remainder is stated as scope, never as equivalence.